

Types of machine learning

Machine learning can be classified into three types of algorithms:

1. Supervised learning
2. Unsupervised learning
3. Reinforcement learning

Supervised learning

Supervised learning, as the name suggests, involves the presence of a supervisor as trainer. Basically, it is a learning process, with which we train or teach the machine using data that is well structured, i.e., data is tagged with correct answers. After that, the machine is provided with new sets of examples (data) so that the supervised learning algorithm analyses the training data (set of training examples) and produces a correct outcome from the structured data.

As an example, consider we have a basket full of different vegetables. The first step is to train the machine about all the different vegetables, one after the other:

- If the shape is cylindrical and the colour is brown, then it will be labelled as a potato.
 - If the shape is like a cylinder and the colour is red or orange-green, then it will be labelled as a tomato.
- The other vegetables are also categorised likewise.

Now, let's suppose that after training the data, you give a separate vegetable, say, a tomato, and ask that it be identified. Since the machine has already learnt some things from previous data, this time it will have to use what it has learnt, sensibly. It will first categorise the vegetable on the basis of its shape and colour, and will validate the vegetable's name as 'Tomato', putting it in the 'Tomato' category. Thus the machine learns certain things from the training data (the basket containing vegetables) and then applies that knowledge to the test data (a new vegetable).

Unsupervised learning

This is the training of the machine using information that is neither classified nor structured, and allowing the algorithm to act on the items without supervision. Here the task of the machine is to group unsorted information according to similarities, patterns and differences without any prior training data. As the name suggests, no training is provided to the machine in this case. Therefore, the machine has to learn by itself from unstructured data. As an example, you may want to find groupings of customer demographics with similar buying habits.

Reinforcement learning

This is an area of machine learning that is related to taking suitable action to maximise rewards in a particular situation, i.e., it is used to find the best possible path for particular circumstances. Reinforcement learning differs from supervised

learning in that, in the latter case, the training data has the answer key with it; so the model is trained with the correct answer itself. However, in reinforcement learning, there is no answer key, and the reinforcement agent decides what to do to perform the given task. In the absence of a training data set, the machine learns from its experience (Figure 1).

We have an agent and a reward, with many hurdles in between. The agent is supposed to find the best possible path to reach the reward. The following example explains this. The objective of a robot is to get the reward, which is a diamond, and avoid the hurdles, which is fire. The robot learns by trying all the possible paths and then choosing the one that gives him the reward with the least hurdles. Each right step will give the robot a reward and each wrong step will take away a reward from it. The total reward will be calculated when it reaches the final reward, which is the diamond.

This article briefly touches on the fundamental concepts of machine learning. I hope it was helpful and will encourage you enough to delve into the topic. **END** 🐧

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